

Xiaoyu Tong

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Education

PhD	University of Amsterdam , Natural Language Processing	Amsterdam, NL
	- Thesis: <i>Dissecting Incongruity: Metaphor and Humor Understanding of Large Language Models</i> - Supervisors: Dr. Ekaterina Shutova, Dr. Martha Lewis - Committee: Prof. Dr. Rens Bod, Dr. Ekaterina Shutova, Dr. Martha Lewis, Prof. Dr. Katja Markert, Dr. Desmond Elliott, Prof. Dr. Khalil Sima'an, Dr. Jelke Bloem, Prof. Dr. Tobias Blanke	July 2026
MSc	University of Amsterdam , Brain and Cognitive Sciences	Amsterdam, NL
	- Thesis: <i>Metaphor Paraphrasing and Word-Sense Disambiguation: Toward a New Approach to Automated Metaphor Processing</i> - Supervisors: Dr. Martha Lewis, Dr. Ekaterina Shutova	Mar 2021
MA	Xiamen University , Linguistics	Xiamen, China
	- Thesis: <i>The Writer's Operation of Extended Metaphor in Meaning Construction</i> - Supervisor: Prof. Zhiying Xin	July 2017
BA	Xiamen University , English	Xiamen, China
	Minor in Chinese Linguistics	July 2014

Experience

Independent Researcher , Postdoctoral Independent Researcher	Remote
- Research on LLMs' cultural alignment in humor processing, collaboration with Dr. Ekaterina Shutova and Dr. Martha Lewis at University of Amsterdam - Research on application of LLMs in speech-language therapy for people with aphasia, collaboration with Dr. Jie Fu at Xingtai Medical College	July 2026 – present 3 months
Faculty of Science, University of Amsterdam , Doctoral Researcher	Amsterdam, NL
Conducted interdisciplinary research on evaluating LLMs' metaphor, humour, pragmatic, and cultural understanding in text and multimodal communication. - Created theory-driven benchmarks and evaluation protocols for testing linguistic, pragmatic, and cultural capabilities in unimodal and multimodal LLMs. - Designed annotation schemes grounded in cognitive and linguistic theories, achieving high reliability (Gwet's AC1 0.84). - Led cross-cultural behavioral studies involving 42K human responses across 4K textual and multimodal stimuli. - Benchmarked frontier open- and closed-source models using prompting and fine-tuning approaches. - Combined statistical analysis, error analysis, and qualitative interpretation to compare human and model behavior.	Sept 2021 – July 2026 4 years 11 months

Institute for Logic, Language and Computation, University of Amsterdam, Research Intern

Amsterdam, NL
May 2020 – Mar 2021
11 months

- Reframed a widely used, theory-driven manual protocol for metaphor identification (MIPVU) as a word-sense disambiguation (WSD) problem, and operationalized this insight in a novel dataset and evaluation framework for automated metaphor interpretation.
- Created a resource that led to the BIG-bench metaphor-understanding benchmark and MUNCH project.

Teaching and Mentoring

- Thesis Mentor, MSc Logic, University of Amsterdam (2023)
- Teaching Assistant, Introduction to the Philosophy of Language, MSc Logic, University of Amsterdam (2022)
- Teaching Assistant, Introduction to Linguistics, BA English, Xiamen University (2015 – 2017)

Service

- Reviewer, ACL Rolling Review (2024 – present)
- Reviewer, *IEEE Transactions on Audio, Speech and Language Processing* (2026)
- Reviewer, *Journal of Cognitive Psychology* (2025)
- Reviewer, NAACL 2024 FigLang Workshop (2024)
- Member, Institute for Logic, Language and Computation (ILLC) PhD Council (2022 – 2023)

Publications

Evaluating LLMs' cultural alignment in humor appreciation Xiaoyu Tong , Ivo Verhoeven, Martha Lewis, Ekaterina Shutova (In Preparation)	present
Dissecting Incongruity: Metaphor and Humor Understanding of Large Language Models Xiaoyu Tong hdl.handle.net/11245.1/0bfb1d70-c5e8-4538-a554-d4c0e2ec0980 (University of Amsterdam)	2026
Hummus: A Dataset of Humorous Multimodal Metaphor Use Xiaoyu Tong , Zhi Zhang, Pia Sommerauer, Martha Lewis, Ekaterina Shutova 10.48550/arXiv.2504.02983 (Conditionally accepted by TACL)	2026
A framework for annotating and modelling intentions behind metaphor use Gianluca Michelli, Xiaoyu Tong (co-first author), Ekaterina Shutova 10.18653/v1/2026.starsem-conference.4 (*SEM 2026)	2026
Metaphor Understanding Challenge Dataset for LLMs Xiaoyu Tong , Rochelle Choenni, Martha Lewis, Ekaterina Shutova 10.18653/v1/2024.acl-long.193 (ACL 2024)	2024
Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models Aarohi Srivastava, Abhinav Rastogi, ... Xiaoyu Tong , ... Ziyi Wu openreview.net/forum?id=uyTL5Bvosj (TMLR, Featured)	2023
Recent advances in neural metaphor processing: A linguistic, cognitive and social perspective Xiaoyu Tong , Ekaterina Shutova, Martha Lewis 10.18653/v1/2021.naacl-main.372 (NAACL 2021)	2021

Projects

Hummus: Dataset and benchmark of humorous multimodal metaphor use github.com/xiaoyuisrain/humorous-multimodal-metaphor-use	2026
MUNCH: Metaphor Understanding Challenge Dataset github.com/xiaoyuisrain/metaphor-understanding-challenge	2024
Co-created dataset of intentions behind metaphor use github.com/GMichelli/intentions-behind-metaphor	2024
BIG-bench metaphor understanding benchmark task github.com/google/BIG-bench/tree/main/bigbench/benchmark_tasks/metaphor_understanding	2021

Skills

Programming and research computing

LLM research

Experimental design, data collection and annotation

Data processing and statistical analysis

Dataset creation and research data management

Research communication and publication

Human languages

Interests

Artificial Intelligence

Linguistics

Cognitive Science